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Abstract

In this paper, we examine the Topologist's Sine Curve, with some discussion on connected and path-connected spaces. We also assume all spaces are Euclidean metric spaces, unless otherwise mentioned.

Definition 1. Let X be a Euclidean space. A **separation** of X is a pair U, V of disjoint nonempty open subsets of X whose union is X.

Definition 2. The space X is said to be **connected** if there does not exists a separation of X.

Definition 3. Given points x and y of the space X, a **path** in X from x to y is a continuous map $f : [a, b] \to X$ of some closed interval of $\mathbb{R}$ into X such that $f(a) = x$ and $f(b) = y$.

Definition 4. A space X is said to be **path-connected** if every pair of points of X can be joined by a path in X.

Proposition 1. Let A, B be spaces with A connected. If $f : A \to B$ is continuous, then $f(A)$ is connected.
Proof. Suppose for contradiction $f(A)$ were not connected. Then, there exists a separation of $f(A) = I \cup J$, with I, J disjoint, nonempty, and open. Then, $f^{-1}(I), f^{-1}(J)$ are disjoint open sets in A with union A, so a separation of A has been found, which contradicts connectedness of A, by assumption.

Proposition 2. Let A be a space, $B \cup C$ a separation of A, and Y a connected subset of A. Then $Y \subset B$ (with $Y \cap C = \emptyset$) or vice-versa.
Proof. It follows that $Y \cap B$ and $Y \cap C$ are open in Y. If both were nonempty, then we get a separation of Y, which contradicts connectedness of Y. Hence, at least one of them must be empty, so Y lies fully in either B or C.

Proposition 3. If a space A is path-connected, then it is connected.
Proof. Suppose for contradiction A is not connected, so there exists separation $B \cup C = A$ of disjoint, nonempty, open subsets B, C of A. By assumption, choose $b \in B$ and $c \in C$. Since A is path-connected, there exists a path $\gamma : [0, 1] \to A$ so that $\gamma(0) = b$ and $\gamma(1) = c$. Since $[0, 1] \subset \mathbb{R}$ is connected, thus $\gamma([0, 1])$ is connected (by proposition 1). Proposition 2 then gives that $\gamma([0, 1])$ lies fully in either B or C, thus implying that there is no path that connects b and c, a contradiction.

We now examine the Topologist's Sine Curve, namely the set $S := \{(x, \sin(\frac{1}{x})) \mid 0 < x \leq 1\} \cup (\{0\} \times [-1, 1])$. This is a classic case of a space that is connected but not path-connected. We first show that S is connected. Put $G := \{(x, \sin(\frac{1}{x}) \mid 0 < x \leq 1)\}$ and $H := \{0\} \times [-1, 1]$. Note G is the image of the connected subset $(0, 1] \subset \mathbb{R}$, so we get that G is connected. Since $\overline{G} = S$ and G is connected, thus S is connected. We omit the proof of the following statement due to time: *the closure of a connected* *set is connected.*

To see that S is not path-connected requires a more elaborate argument, which we will omit here due to time. Individually, the vertical bar and sine curve components are path-connected. Intuitively, if, given a point on the sine curve component and a point on the vertical bar component, there exists a continuous path in S joining them, then we would require an "infinite amount of time" to travel along the path to reach one point from the other.

However, there exists a partial converse to proposition 3, which we will now discuss.

Consider now the following lemma and theorem.

Lemma. *If Ω is closed and open in a connected space X, then $\Omega = X$ or $\Omega = \emptyset$.*

Proof. If $\Omega = \emptyset$, we are done. Now let $\Omega \neq \emptyset$ with $\Omega \subsetneq X$. By assumption, we have $X \setminus \Omega$ is also closed and open. Write $X = \Omega \cup (X \setminus \Omega)$. Thus, $\Omega, (X \setminus \Omega)$ are open, disjoint, nonempty subsets of X, and thus they constitute a separation of X, contradicting connectedness of X. $\square$

Theorem. *Let $X \subseteq \mathbb{R}^n$ be open. If X is connected, then X is path-connected.*

Proof. Fix $x \in X$. Define Ω to be the set of all $y \in X$ such that there exists a continuous path connecting x and y. Clearly, $x \in \Omega$ (by the trivial path), so $\Omega \neq \emptyset$. If we show that Ω is closed and open in X, then by the above lemma, we may conclude that that $\Omega = X$; this conclusion then gives that, since $x \in X$ was arbitrary, there exists a path between any two points of X fully contained in X. Now, we show Ω is open in X. Take $y \in \Omega$ and show that y is an interior point of Ω. Since $y \in \Omega \subset X$, thus since X is open, y is an interior point of X, so there exists a $\delta > 0$ such that $\overline{B_\delta(y)} \subset X$. Now we show $\overline{B_\delta(y)} \subset \Omega$, and we do this by showing each $z \in \overline{B_\delta(y)}$ satisfies $z \in \Omega$. Take $z \in \overline{B_\delta(y)}$ arbitrarily. We now construct a continuous path connecting x and z as follows. Since $y \in \Omega$, let γ be a continuous path connecting x and y. By the choice of z, we have $d(y, z) \leq \delta$ and so construct a straight-line path $\Gamma : [0, 1] \to \overline{B_\delta(y)}$ with $\Gamma(t) = (1 - t)y + tz$. Thus, for any $t \in [0, 1]$ we get a point $w \in \Gamma([0, 1])$ with $|w| = |(1 - t)y + tz| \leq |(1 - t)y| + |tz| \leq |1 - t||y| + |t||z|$. Define a Euclidean coordinate system $\mathbb{E}^n$ with y as the origin in $\mathbb{E}^n$ (and $\mathbb{E} \cong \mathbb{R}$) thus giving $|y| = 0$. Our inequality chain is thus at most $|t||z| \leq \delta$, by the choice of z and definition of t. Thus we have that the straight-line path Γ from y to z is contained in $\overline{B_\delta(y)}$, and trivially, Γ is continuous in its parameter, t. Now construct a path $H := \gamma \cup \Gamma$. Assuming, WLOG, γ has domain $[0, 1]$, we have $\gamma(1) = \Gamma(0)$ and so this concatenation H of continuous paths γ and Γ gives a continuous path connecting x and z. Thus, $z \in \Omega$ and so since z was arbitrary in $\overline{B_\delta(y)}$, thus $\overline{B_\delta(y)} \subset \Omega$, so y is an interior point of Ω. Since y was arbitrary in Ω, thus Ω is open.

It remains to show that Ω is closed. We may instead show $X \setminus \Omega$ is open. Take

$y \in X \setminus \Omega$ arbitrarily and show y is an interior point of $X \setminus \Omega$. By assumption, there does not exist a continuous path connecting x and y. Since y is an interior point of X, there exists an $\epsilon > 0$ such that $\overline{B_\epsilon(y)} \subset X$. We show that $z \in \overline{B_\epsilon(y)} \subset X \setminus \Omega \subset X$. Suppose for contradiction there is a continuous path connecting x and z, i.e. $z \in \Omega$. Using a construction similar to the above paragraph, it follows that there is a continuous path connecting x and y, thus contradicting the assumption $y \in X \setminus \Omega$. Therefore, $X \setminus \Omega$ is open and so Ω is closed. By the above lemma, we conclude $\Omega = X$. $\square$